Simulation Based Room Security and Home Automation System Using Digital Logic Gates

Pratik Tanty
Department of Electronics and Communication Engineering
Birla Institute of Technology, Mesra
Ranchi, Jharkhand, India
btech10211.25@bitmesra.ac.in

Piyush Pranjal
Department of Electronics and Communication Engineering
Birla Institute of Technology, Mesra
Ranchi, Jharkhand, India
btech10212.25@bitmesra.ac.in

P Sampat Ganesh
Department of Electronics and Communication Engineering
Birla Institute of Technology, Mesra
Ranchi, Jharkhand, India
btech10221.25@bitmesra.ac.in

Kshitij Kumar
Department of Electronics and Communication Engineering
Birla Institute of Technology, Mesra
Ranchi, Jharkhand, India
btech10224.25@bitmesra.ac.in

***Abstract*—This paper presents the design and simulation of a dual purpose digital logic circuit that integrates password based security with intelligent home automation. The system was implemented using fundamental logic gates (XNOR, AND, OR, NOT) in NI Multisim without microcontrollers, demonstrating core digital design principles for first-year electronics students. The circuit performs three key functions: password verification through 3-bit comparison, sensor based energy management, and automated control of room appliances based on environmental sensors. The security module employs XNOR gates as bit comparators to validate user input against stored passwords, with LED indicators providing visual feedback. Upon successful authentication, the automation subsystem is conditionally activated on the basis of the selected operational mode. The simulation results demonstrate successful password validation with 100accuracy, proper mode-switching behavior, and reliable sensor-based appliance control. The system finds practical applications in the smart home, hotel room management, office buildings, and educational laboratories to teach the fundamentals of digital logic.**

***Index Terms*—digital logic design, home security system, password authentication, XNOR comparator, sensor-based automation, energy management, Multisim simulation**

I. Introduction

Integration of security and automation in residential and commercial spaces has become increasingly important with increasing concerns about energy efficiency and unauthorized access. Traditional home security systems operate independently of automation controls, leading to potential security vulnerabilities where intruders could manipulate smart home features.

Modern smart home solutions typically employ microcontrollers, wireless communication modules, and sophisticated software algorithms. While these approaches offer flexibility and advanced features, they abstract away the fundamental digital logic principles that underpin all electronic control systems. For first-year electronics engineering students, understanding how basic logic gates can create complex decision-making circuits is crucial before advancing to programmed microcontroller implementations.

This project aims to design a room security and automation system using only combinational and sequential logic circuits. The educational objective is to demonstrate that intelligent systems can be built from fundamental components—XNOR gates, AND gates, OR gates without requiring software programming. The design incorporates three core features: password-based access control, operational mode selection, and sensor driven appliance management.

The system addresses a common real-world scenario: a room or facility that requires both security and energy efficiency. Unauthorized users are denied access, and even authorized users can only activate automation features when appropriate.

Unlike microcontroller-based projects that hide circuit behavior inside program code, this pure logic implementation makes every decision visible through gate connections and signal flow. Students can trace how a 3-bit password transforms into an access decision, how mode selection gates the automation path, and how sensor signals combine with authentication to control real outputs. This serves as an invaluable educational tool to understand the principles of digital system design that remain relevant regardless of the implementation technology.

II. Background and Related Work

A. Digital Security Systems

Electronic security systems have evolved from mechanical locks to sophisticated digital access control. Early digital locks used simple relay-based circuits with fixed code switches. Modern systems employ keypad interfaces, RFID cards, or biometric sensors, typically managed by microcontrollers. However, the core principle comparing user input against stored credentials remains fundamentally a bit-comparison operation suitable for implementation with XNOR logic gates. [3].

B. Home Automation Technologies

Home automation systems control lighting, climate, and appliances based on various inputs including manual commands, timers, and environmental sensors. Commercial products like smart thermostats and lighting controllers integrate multiple sensors with cloud connectivity and machine learning algorithms. [1].

C. Logic Gate Implementation

XNOR gates are well-suited for equality checking because they output a high level only when both inputs are identical. By combining several XNOR outputs using an AND gate, multi-bit comparisons can be achieved without the need for a processor.

D. Research Gap and Contribution

While numerous student projects implement either digital locks or sensor-based automation independently, few demonstrate the integration of both subsystems with conditional enabling logic. Most educational resources progress directly from basic gate examples to microcontroller programming, skipping the intermediate complexity level where multiple logic functions interact. This project fills that gap by presenting a complete system that requires students to understand bit comparison and multi-input control logic simultaneously.

III. System Design

A. Overall Architecture

The system architecture consists of interconnected modules operating in a hierarchical structure. At the input level, user interface elements include three binary switches for password entry, a push-button for password verification and simulated sensor inputs for motion detection and temperature monitoring.

The authentication module output serves as a master enable signal for the automation subsystem. A secondary enable path incorporates the mode selection, creating a two-level interlock system. Only when both authentication succeeds AND checking mode is active, the automation logic process sensor works. The output stage consists of LED indicators for system status (access granted/denied) and appliance control signals (light and air conditioning activation).

B. Password Authentication Module

The security module implements a 3-bit password comparison system using XNOR gates as the fundamental building blocks. Three input switches represent the user-entered password, while three stored values are hardwired as the correct password. In the Multisim implementation, stored values are set using fixed logic levels.

Each bit position uses one XNOR gate to compare input against the stored value. The XNOR operation produces logic 1 output only when both inputs are identical that is, either both 0 or both 1. This property makes XNOR ideal for equality checking. The three XNOR outputs feed into a 3-input AND gate, which produces logic 1 only when all three bit comparisons succeed simultaneously.

Two LED indicators provide immediate visual feedback. A green LED connects to the Access Granted signal, illuminating when the password matches. A red LED connects to the inverted signal (NOT gate applied to Access Granted), illuminating when password comparison fails. This ensures exactly one LED is always lit, providing unambiguous status information.

The 3-bit password provides 8 possible combinations, which is sufficient for a demonstration system. Real-world implementations typically use 16 to 64 bits for adequate security.

C. Day/Night Mode Control

The operational mode selector provides energy management functionality by enabling or disabling automation features based on the users' requirements. A single-pole toggle switch serves as the mode input, with logic 1 representing day mode (automation enabled) and logic 0 representing night mode (automation disabled).

The mode signal combines with the authentication signal through an AND gate before reaching the automation module. This creates a dual-condition requirement: automation activates only when (Access Granted = 1) AND (Day Mode = 1). The logic equation is:

Automation Enable = Access Granted · Day Mode

This design allows authorized users to control energy consumption patterns. During daytime (day mode active), the system responds to motion and temperature sensors to manage lighting and climate control. During nighttime (night mode active), automation remains disabled regardless of sensor states, preventing lights from activating due to incidental movement during sleep hours and avoiding unnecessary air conditioning operation.

The mode selection adds practical utility beyond basic demonstration. Hotel rooms could default to night mode when guests indicate they're sleeping. Office spaces could automatically switch modes based on business hours. Residential users gain manual control over when smart features operate, balancing convenience with energy conservation.

D. Appliance Control Logic

Two independent automation paths control room appliances: lighting control and climate control. Each path implements its own three-input AND gate combining sensor input, authentication status, and mode selection.

For lighting control, the logic equation is:

Light ON = Motion Detected · Access Granted · Day Mode

The light (represented by a yellow LED in simulation) activates only when all three conditions are simultaneously true: movement is detected, the correct password has been entered, and day mode is selected. If any condition fails, the light remains off. This prevents lights from activating when the room is unoccupied, when unauthorized users are present, or during designated rest periods.

For climate control, the logic equation is:

AC ON = Temperature High · Access Granted · Day Mode

The air conditioner (represented by a blue LED in simulation) activates when temperature exceeds the threshold, authentication succeeds, and day mode is active. This prevents energy waste cooling empty or unauthorized rooms, and respects user preference to disable cooling during night hours when occupants might prefer different thermal conditions.

E. Component Specifications

Table I lists all components used in the Multisim simulation. I

TABLE I
CIRCUIT COMPONENTS

Component	Quantity	Function
XNOR Gate	3 gates	Password bit comparison
AND Gate	3 ICs	Authentication, mode, appliance control
OR Gate	1 IC	(if needed for expansion)
NOT Gate	1 IC	Red LED inverter
Toggle Switch	6	Password inputs (3) + mode (1)
LED Indicator	4	Green, Red, Yellow, Blue

IV. Simulation and Results

A. Simulation Setup

The complete circuit was designed and simulated in NI Multisim 14.2 educational edition. Virtual instruments including logic probes.

Four test scenarios were designed to validate system functionality: correct password entry with day mode, incorrect password rejection and day versus night mode comparison. Each test scenario exercised specific circuit paths to ensure proper operation under different input conditions.

B. Test Case 1: Correct Password Entry (Day Mode)

The stored password was configured as binary 101. Input switches were set to match this pattern. The mode switch was set to day mode (logic 1). Upon verification button press, the following observations were made:

- All three XNOR gates produced logic 1 outputs - The 3-input AND gate output transitioned to logic 1 - Green LED illuminated immediately - Red LED extinguished - Automation Enable signal became active - Motion sensor input (set to logic 1) triggered yellow LED activation - Temperature sensor input (set to logic 1) triggered blue LED activation

This test confirmed that correct authentication in day mode enables full system functionality. Both sensor-driven automation paths operated as designed, with appliances responding appropriately to environmental conditions.

C. Test Case 2: Incorrect Password Entry

Input switches were deliberately set to binary 111, differing from the correct stored password. All other conditions remained identical to Test Case 1.

Observations: - Two XNOR gates (positions 0,2) produced logic 1 - One XNOR gate (position 1) produced logic 0 - The 3-input AND gate output remained at logic 0 - Red LED illuminated, green LED extinguished - Automation Enable signal remained inactive (logic 0) - Despite motion and temperature sensors being active, both yellow and blue LEDs remained off - Appliance control circuits correctly inhibited outputs

This test confirmed the security interlock functions properly. A single incorrect bit prevents authentication, and failed authentication blocks all automation features regardless of sensor states.

D. Test Case 3: Day Mode vs Night Mode Comparison

The correct password (101) was entered, ensuring Access Granted = 1. Motion and temperature sensors were activated (both logic 1). The mode switch was then toggled between day and night positions.

Day Mode (Mode = 1): - Green LED on (authentication successful) - Yellow LED on (motion detected, automation enabled) - Blue LED on (temperature high, automation enabled) - Both appliances operating normally

Night Mode (Mode = 0): - Green LED remained on (authentication still valid) - Yellow LED off (motion detected but automation disabled) - Blue LED off (temperature high but automation disabled) - No appliances activated despite active sensors

This test demonstrated the mode selection functionality. Authorized users retain authentication status in both modes, but automation responds only in day mode. The system correctly interprets that night mode represents a user preference to disable automatic responses, conserving energy during rest periods.

V. Discussion

A. Design Validation

The simulation results comprehensively validate the system design. All functional requirements were met: password authentication correctly accepts only the exact stored pattern, mode selection appropriately enables or disables automation, and sensor-driven appliance control operates conditionally based on authentication and mode status. The hierarchical interlock structure functions as intended, with security taking priority over all other operations.

B. Educational Value

From a pedagogical perspective, this project successfully demonstrates several fundamental concepts in digital design. Students observe how XNOR gates implement equality comparison, how cascaded AND gates enforce multiple simultaneous conditions, and how signal routing creates hierarchical control structures. The visibility of signal flow through discrete components provides insights often obscured in microcontroller implementations where logic exists as software abstractions.

C. Practical Considerations

While suitable for educational purposes and proof-of-concept demonstration, several practical limitations constrain real-world deployment. The fixed 3-bit password offers insufficient security, vulnerable to exhaustive search (only 8 attempts required). The hardwired password cannot be changed without circuit modification. The absence of attempt limiting allows unlimited password trials.

D. Energy Management Effectiveness

The day/night mode feature successfully demonstrates the principle of conditional automation for energy conservation. By disabling sensor responses during designated periods, the system prevents unnecessary appliance operation. In a residential scenario, night mode would eliminate lights activating due to movement during sleep and avoid overcooling when reduced activity generates less heat.

VI. Applications

A. Residential Smart Homes

The system architecture directly applies to residential security and home automation. Homeowners could deploy the password-based access control at entry points, with automation controlling interior lighting and climate. The day/night mode mirrors real homeowner preferences—active automation during waking hours, disabled during sleep.

B. Hotel Room Management

Hotels face significant challenges with energy waste when guests leave appliances running in unoccupied rooms. This system concept addresses that problem by requiring authentication (via key card encoding) before enabling room features. The day/night mode could interface with hotel ”Do Not Disturb” indicators—automation disabled when privacy is requested.

C. Office and Commercial Buildings

Office spaces could implement this design for individual rooms or zones, with employee access codes enabling both entry and automation. The day/night mode maps naturally to business hours—full automation during 9-5 operations, minimal energy use overnight. Meeting rooms particularly benefit from occupancy-based lighting and climate control, avoiding the common problem of running HVAC in empty conference spaces.

D. Server Room and Data Center Security

Server rooms require stringent access control combined with critical environmental monitoring. Unauthorized access could enable data theft or sabotage, while improper climate control risks hardware damage. This system’s integration of authentication with environmental control aligns well with data center requirements. Extensions could add humidity sensing, door position monitoring, and redundant cooling systems.

E. Industrial Control Applications

Manufacturing facilities employ similar interlocked safety systems where machine operation requires authentication and appropriate environmental conditions. The principle of multi-condition enabling—authentication, mode selection, and sensor verification—mirrors industrial safety protocols preventing unauthorized machine activation or operation under hazardous conditions.

VII. Conclusion

This paper presented a comprehensive room security and home automation system implemented entirely using fundamental digital logic gates without microcontrollers or programming. The design successfully integrated password-based authentication, operational mode selection, and sensor-driven appliance control into a cohesive system demonstrating both security and energy management principles.

Simulation results validated all functional requirements. Password authentication achieved 100accuracy, correctly accepting matching 3-bit patterns while rejecting all incorrect attempts. The day/night mode feature appropriately enabled or disabled automation based on user preference. Sensor-driven appliance control functioned correctly under all tested conditions, with the hierarchical interlock structure ensuring security always takes priority over convenience features.

The educational value of this pure logic implementation lies in its transparency. Every decision—from bit-by-bit password comparison through conditional appliance control—occurs through visible gate connections that students can trace and verify. This hands-on understanding of how simple logic elements combine to create intelligent behavior provides essential foundations for advanced digital design.

Future enhancements could extend system capabilities in several directions. Adding multiple password storage with user identification would demonstrate memory concepts. Implementing attempt counting and timeout features would illustrate sequential circuit design using flip-flops and counters.

This project successfully achieved its objective of creating a functional, integrated security and automation system using only digital logic fundamentals, providing first-year electronics students with valuable experience in systematic digital design methodology.

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